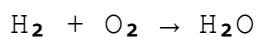


Kai Ash

~20 minutes · Show all work

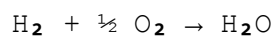
WORKED EXAMPLE — THE FRACTION MOVE

Balance combustion equations by saving O_2 for last. If you need an odd number of O atoms, write a fractional O_2 coefficient, then multiply every coefficient by the denominator to clear the fraction.

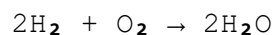


Step 1. Balance H → 1 H_2O gives 2 H ✓

Step 2. Balance O → right side has 1 O, but O_2 supplies 2 → write $\frac{1}{2} O_2$:



Step 3. Multiply every coefficient $\times 2$ to remove the fraction:



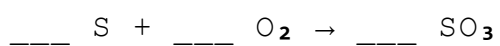
Check: H = 4 = 4 ✓ O = 2 = 2 ✓

YOUR TURN

PROBLEM 1

Practice — same move

Balance the equation below using the fraction-then-multiply technique.



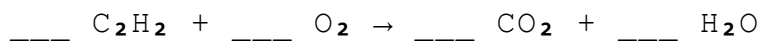
work

Balanced:

PROBLEM 2

Build up

Balance the combustion of acetylene. Save O_2 for last.

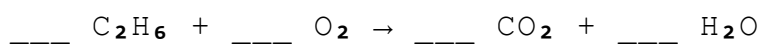


work

Balanced:

PROBLEM 3

Full combustion

Balance the combustion of ethane. Save O₂ for last.

work

Balanced: _____